

Practical-5

- A producer-consumer program was created using an anonymous pipe.
- The parent process acted as the producer and generated the values 10, 20, 30, 40 and 50.
- The child process acted as the consumer and successfully received the same values through the pipe.
- The program measured the communication time using clock().
- The output showed that the data produced by the parent was successfully consumed by the child process.
- The communication time displayed in the executed output was 0.000104 seconds for the producer and 0.000201 seconds for the consumer.

Screenshot of the Output

```
shravya@Shravya:~$ nano producer.c
shravya@Shravya:~$ gcc producer.c
shravya@Shravya:~$ ./a.out
producer
produced 10
produced 20
produced 30
produced 40
produced 50
time taken is 0.000104
consumer
Consumed 10
Consumed 20
Consumed 30
Consumed 40
Consumed 50
time taken is 0.000201
shravya@Shravya:~$ |
```

Screenshot of the Output (practical5.c)

```
shravya@Shravya:~$ nano practical5.c
shravya@Shravya:~$ gcc practical5.c
shravya@Shravya:~$ ./a.out
--w---x--T 1 shravya shravya      0 Aug  3 15:01 c2
-rw-r--r-- 1 shravya shravya      0 Aug 24 05:47 copy.txt
-rw-r--r-- 1 shravya shravya     65 Aug 27 04:42 exe.c
-rw-r--r-- 1 shravya shravya    249 Aug 27 04:52 exe1.c
-rw----- 1 shravya shravya     54 Aug 27 05:13 exe2.c.save
-rw-r--r-- 1 shravya shravya     74 Aug 11 04:55 exec1.c
-rw-r--r-- 1 shravya shravya    111 Aug 11 04:59 exec2.c
-rw-r--r-- 1 shravya shravya    132 Aug 11 05:35 exec3.c
-rw-r--r-- 1 shravya shravya     85 Aug 11 05:12 exexc3.c
-rw-r--r-- 1 shravya shravya    392 Sep  1 05:42 fcfs.c
-rw-r--r-- 1 shravya shravya    188 Aug  1 09:40 fork.c
-rw-r--r-- 1 shravya shravya    387 Aug  4 05:44 fork1.c
-rw-r--r-- 1 shravya shravya   4444 Aug 24 16:13 fork3.c
-rw-r--r-- 1 shravya shravya    279 Sep 15 05:58 malloc.c
-rwxr-xr-x 1 shravya shravya 26664 Sep 22 05:18 malloc_asan
-rw-r--r-- 1 shravya shravya    318 Aug 27 05:49 namedpipeex.c
-rw-r--r-- 1 shravya shravya    347 Aug 24 05:43 open2.c
-rw-r--r-- 1 shravya shravya    246 Aug  8 09:32 pipe.c
-rw-r--r-- 1 shravya shravya    303 Aug  8 10:09 pipe1.c
-rw-r--r-- 1 shravya shravya    587 Aug  8 09:32 pipe2.c
-rw-r--r-- 1 shravya shravya    230 Aug 11 05:59 pipeex.c
-rw-r--r-- 1 shravya shravya    235 Aug 11 06:01 pipex.c
-rw-r--r-- 1 shravya shravya    294 Aug 24 05:26 practical
-rw-r--r-- 1 shravya shravya    293 Aug 24 05:30 practical.c
-rw-r--r-- 1 shravya shravya     26 Aug 24 05:49 practical.txt
-rwxr-xr-x 1 shravya shravya 16176 Aug 24 05:56 practical1
-rw-r--r-- 1 shravya shravya    303 Aug 24 05:55 practical1.c
-rw-r--r-- 1 shravya shravya    246 Sep 24 03:43 practical5.c
-rw-r--r-- 1 shravya shravya    338 Sep  1 05:30 process1.c
-rw-r--r-- 1 shravya shravya    255 Sep  1 05:24 process2.c
-rw-r--r-- 1 shravya shravya    588 Sep 24 03:31 producer.c
-rw-r--r-- 1 shravya shravya      0 Jul 28 05:46 public.txt
-rw-r--r-- 1 shravya shravya     84 Aug  3 06:49 read1.c
-rw-r--r-- 1 shravya shravya    277 Aug  3 08:29 readwrite.c
-rw-r--r-- 1 shravya shravya    225 Aug  3 15:00 remove.c
---x--x--x 1 shravya shravya      0 Jul 28 05:47 script.sh
-rw-r--r-- 1 shravya shravya    426 Sep 24 03:13 waitx.c
-rw-r--r-- 1 shravya shravya    219 Aug  3 08:29 write.c
-rw-r--r-- 1 shravya shravya    219 Aug  3 16:14 write.c.save
-rw----- 1 shravya shravya     88 Aug  3 10:16 write2.c.save
-rw-r--r-- 1 shravya shravya    293 Sep 24 03:21 zombie.c
shravya@Shravya:~$ |
```